13. Write the python program to implement Minimax algorithm for gaming

Aim: To implement the Minimax algorithm for selecting the optimal move in a game.

Procedure:

1. Define the possible game states.
2. Identify the maximizing and minimizing players.
3. Generate the possible moves.
4. Evaluate the terminal states using scores.
5. Apply Minimax recursively to choose the best score.
6. Display the optimal move and its score.

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```
def minimax(depth, index, maximizing, scores, max_depth):
    if depth == max_depth:
        return scores[index]

    if maximizing:
        return max(
            minimax(depth + 1, index * 2, False,
                    scores, max_depth),
            minimax(depth + 1, index * 2 + 1, False,
                    scores, max_depth)
        )
    else:
        return min(
            minimax(depth + 1, index * 2, True,
                    scores, max_depth),
            minimax(depth + 1, index * 2 + 1, True,
                    scores, max_depth)
        )

scores = [3, 5, 2, 9, 12, 5, 23, 23]
depth = 3

result = minimax(0, 0, True, scores, depth)
print("Optimal Score:", result)
```

Output:

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```
Python 3.14.7 (tags/v3.14.7:823
AMD64)] on win32
Enter "help" below or click "He
>>>
========== RESTART: C:\Use
Optimal Score: 12
>>>
```

Result: Thus, the Minimax algorithm was implemented successfully to determine the optimal game score.